

International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Mid-Term Examination, Spring 2025
Course Code: CSE 3521 Course Title: Computer Architecture
Total marks: 30 Time: 1.5 hours

- | | | | |
|---|-----------|----------|---------|
| 1. a) Define CPI and explain how it affects CPU performance. Explain with example | Mark
3 | CLO
2 | DL
2 |
| ↓ b) Why is MIPS not always a reliable performance measure? Explain with example | 3 | 2 | 2 |
| 1. c) Different instruction sequences can yield different execution times for the same task. Given two compilers A and B, both targeting the same hardware: | 4 | 2 | 3 |

Class	CPI	A Count	B Count
A	1	5	4
B	2	2	3
C	3	1	2

Assuming a 2.5 GHz clock, compute which compiler produces a faster code and justify your answer with proper reasoning.

- OR
- Consider the following two processor designs for the same ISA:
Processor A: Clock Rate = 2.5 GHz, Average CPI = 1.6
Processor B: Clock Rate = 2.0 GHz, Average CPI = 1.2
 A program takes 1.5 billion instructions to execute.
 a. Compute the execution time on each processor.
 b. Which processor is faster and by what percentage?
 c. If performance is critical, which design would you recommend?

$$\begin{aligned} \text{Execution Time} &= \frac{\text{Instructions}}{\text{Clock Rate} \times \text{CPI}} \\ \text{For Processor A: } &= \frac{1.5 \times 10^9}{2.5 \times 10^9 \times 1.6} = 0.375 \text{ s} \\ \text{For Processor B: } &= \frac{1.5 \times 10^9}{2.0 \times 10^9 \times 1.2} = 0.625 \text{ s} \end{aligned}$$

2. a) Translate the following MIPS instructions into machine code:
 (i) addi \$t1, \$t2, 15
 (ii) beq \$s1, \$s2, -8
 (iii) lw \$t0, 32(\$sp)

- OR
- Translate the following machine code into MIPS instructions:
 001100 01011 01010 0000000011110000
 101011 10001 10000 1111111111101100
 000101 00101 00110 0000000000000100

- | | | | |
|--|---|---|---|
| 2. b) Describe the purpose and structure of an instruction format. How it aligns with design principles. | 4 | 1 | 2 |
| 3. a) A computer system is using the IEEE 754 format to store floating-point numbers. Why is normalization important in this format? Show how the number -5.75 is normalized and represented in single precision. | 6 | 1 | 3 |
| 3. b) Why refined versions of multiplication hardware reduce its size and components? Explain with examples. | 4 | 1 | 2 |

International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE, Spring-2025, Midterm Exam

Course Code: CSE-3529, Title: System Analysis and Design

Full Marks: 30, Time: 1 Hour 30 Minutes

Answer all of the following questions

- | | | Marks | |
|------------|---|-------|-----|
| 1. | a) Define information. Discuss the key qualities that make information valuable and effective in decision-making. | 5 | CO1 |
| | b) A software development team is working on a complex project with high-risk factors and frequent changes in requirements. The project manager decides to divide the development into smaller, manageable phases, each with risk assessment, prototyping, and client feedback. After each phase, the team reviews the progress, adjusts plans, and moves on to the next iteration.

What software development model is being used in this scenario, and how does it address the challenges of risk management and iterative development? | 5 | CO3 |
| 2. | a) What is a system? Discuss different types of information systems and their roles in organizations. | 5 | CO1 |
| | b) What is the Systems Development Life Cycle (SDLC), and how does it help in building an information system? Explain the key phases of the SDLC and provide an example of how each phase might be applied in the development of an e-commerce platform. | 5 | CO1 |
| OR (of 2b) | | | |
| | Which SDLC model would you adopt for developing an e-commerce platform that requires small and there are no uncertain requirements? Justify your answer reasoning with the advantages and disadvantages of your model. | 5 | |
| 3. | a) Discuss both internal and external factors that affect system project. | 5 | CO1 |
| | b) Explain the role of SWOT analysis in the strategic planning process for IT projects. How does a SWOT analysis help IT managers in identifying strengths, weaknesses, opportunities, and threats? Provide examples of how each element can impact IT project decisions | 5 | CO3 |
| OR (of 3b) | | | |
| | Describe the steps to conduct the economic feasibility analysis with proper example. | 5 | |

International Islamic University Chittagong
Department of Computer Science and Engineering (CSE)

B.Sc. Engineering in CSE
Midterm Examination, Spring 2025

Course Code: EEE-2421

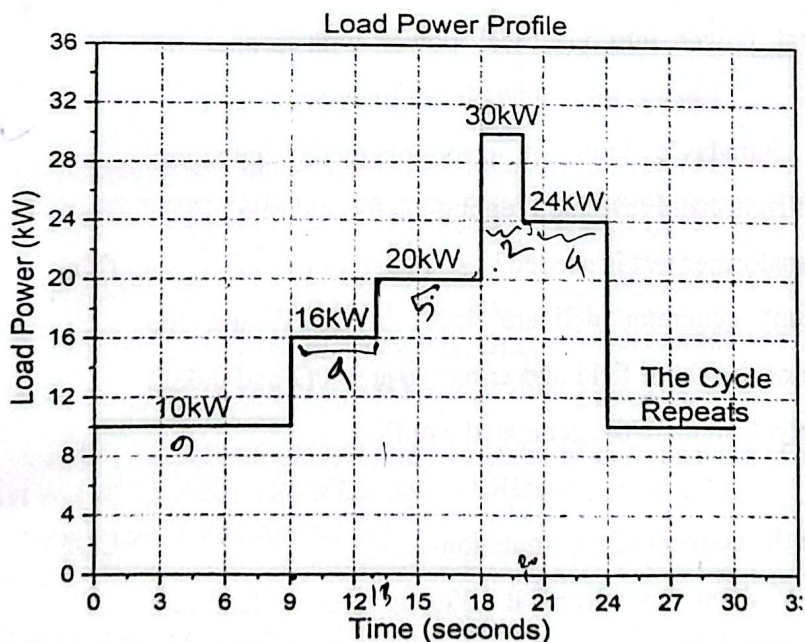
Time: 1.5 hours

Course Title: Electrical Drives and Instrumentation

Full Marks: 30

Answer all the questions. The figures in the right-hand margin indicate full marks
Course Outcome (COs) and Bloom's Level (BL) are mentioned in additional Columns

- 1(a) i. List the different types of motor duty classes and explain any two of them with timing diagram. CO1 Ap 5
- ii. The magnetic flux linked with a coil having 250 turns is changed from 1.4 Wb to 2 Wb in 0.45 seconds. Calculate the induced e.m.f in the coil. 33.33
- 1(b) The following load power profile was measured for a constant speed load. Choose an appropriately sized motor to drive this Load. Assuming a service factor of 30%. You do not need to consider starting performance. CO3 Ap 5



- 2(a) i. What is transformer? Draw the equivalent circuit of transformer with proper notation. CO1 R 5
- ii. Describe the core losses in a transformer.

2 (b)

i. Classify the transformer test.

CO3 An 1+4

ii. In no-load test of single-phase transformer, the following test data were obtained:

Primary voltage: 230V; Secondary voltage: 115V

Power Input: 40W; Primary Current: 0.5A

Find the following:

(i) the turns ratio 2

(ii) the magnetizing component of no-load current 0.468%

(iii) its working component 0.173%

(iv) the iron loss

$$P = 0.6$$

OR

2 (b)

i. Distinguish between the transformer test.

CO1 An 1+4

ii. A 5 kVA 200/1000 V, 50 Hz, single-phase transformer gave the following test results:

O.C. Test (L.V. Side): 200 V, 1.2 A, 90 W

S.C. Test (H.V. Side): 50 V, 5A, 110 W

Calculate the parameters of the equivalent circuit referred to the L.V. side.

3 (a)

i. Describe the operation of DC machine that converts mechanical power into electrical power with proper net sketch.

CO1 R 5

ii. State Faraday's laws of electromagnetic induction, Fleming's right-hand rule and Fleming's left-hand rule.

$V = I_a R_a$

3 (b)

i. Why transformer test is needed?

CO3 An 1+4

ii. A shunt generator delivers 455A at 230V and the resistance of the shunt field and armature are 40Ω and 0.03Ω respectively. Calculate the generated e.m.f.

$$V = I_a R_a$$

$$950 - I$$

OR

3 (b)

i. Analyze the concept of commutation.

CO3 An 1+4

ii. A 255-V shunt motor runs at 1000 r.p.m at no-load and takes 9A. The total armature and shunt field resistances are respectively 0.2Ω and 255Ω . Calculate the speed when loaded and taking 50A. Assume the flux to be constant.

International Islamic University Chittagong
Department of Computer Science and Engineering
Program: B.Sc. in CSE: Semester: 5th
Mid-Term Examination, Spring-2025

Course Code: CSE 3523 Course Title: Microprocessor, Microcontroller and Embedded System
Time: 1 hour 30 minutes Full Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

Questions

1. ✓ a) Describe the physical address calculation in the 8086 microprocessor. 2
Include how segment and offset registers are used.
Find the physical address of 1000:2000. CO1

1. ✓ b) Write short notes on the following: 3
 - i) CISC
 - ii) RISC CO1
 - iii) ALU

1. ✓ c) Explain in detail the steps involved in the execution cycle of a  microprocessor. 5
CO2

2. ✓ a) Draw the internal block diagram of the 8086 microprocessor. 4
CO1

2. ✓ b) what are the addressing modes used in the following Instructions? If the instructions are incorrect, then explain the reasons and make them correct 2
CO1
 - i. MOV AL, BL
 - ii. MOV AL, BX
 - iii. MOV AH, [BX]
 - iv. MOV CL, 3A8H

2. ✓ c) Write a short assembly program to add two 16-bit hexadecimal numbers 20F1H and 3C72H using 8-bit addition. Also, explain how the ADC instruction works in this context. 4
CO2

2. c) Find the value of C, A, P, Z, S, O FLAGS after execution of this instruction. 4
CO1

OR,

```
MOV AL, 3FH
MOV BL, 2AH
ADD AL, BL
```

3. a) Write about the advantages and disadvantages of low-level language. CO2 2
3. b) Write an assembly language program to set the rightmost 3-bits, to clear the leftmost 2-bits, and to Invert the bit position 5, 7 & 8th of register value AX=FF0EH. CO2 4
3. c) Given, AX=8877 H and BX=79ABH. Interchange the value of AX and BX using PUSH and POP Operation. Draw the Table of AX, BX and SP Value after every instruction. Take the initial address, SP= 2009H. CO2 4

OR

3. c) What is the operation of the registers in the 8086 microprocessor during 8-bit and 16-bit multiplication? Write assembly language for the equation $y=6 \times 8 \times 10H$. CO2 4

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INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science & Engineering (CSE)

BSc in CSE, Mid-Term Examination, Spring 2025

Course Code: CSE-3527 Course Title: Compiler

Full Marks: 30 Time: 1 Hour 30 minutes

[Answer all the following questions. Some questions may have an option.
(Figures at right margin illustrate the marks (M), course objectives (COs), and bloom's taxonomy (DL))]

No.	Questions	M	COs	DL
1. a)	Define the compiler with a block diagram that illustrating its front-end and back-end phases. Elucidate the analysis of the source program with its compiling phases.	4 (2+2)	CO1	U
Or				
	How is an executable target program generated from a given source program? Explain it using a language processing system with an appropriate diagram. What basic tools are needed for compiler construction in a language processing system?	4 (1+2+1)	CO1	An
b)	Translate the following C programming statement step by step through the compiler phases. Draw the diagram of the compiler's phases with output translating annotations at each phase and generate the symbol table for the following statement. $a = x + y * r - w / 4;$ [where the values of a , r , and w are fractional]	6 (2+3+1)	CO1	An
2. a)	Write is the purpose of context-free grammar (CFG) in compiler design. Define and emphasize the concept of ambiguity in CFG. How does it impact a compiler design? Consider the following CFG G_1 : $\begin{aligned} Exp &\rightarrow Exp Op Exp \\ Exp &\rightarrow (Exp) \\ Op &\rightarrow + - * / \\ Exp &\rightarrow int \end{aligned}$ Whether the above grammar is ambiguous or not. Justify your answer using its input string " $int * (int + int)$ ". Construct the syntax tree for the input string " $int + int * (int / int)$ ".	6 (3+3)	CO2	Ap
Or				
	Define synthesized attributes? Consider the following CFG G_2 : and G_3 $\begin{aligned} G_2: \quad S &\rightarrow Ab aB \\ A &\rightarrow aA \epsilon \\ B &\rightarrow Bb \epsilon \end{aligned} \quad \begin{aligned} G_3: \quad Seq &\rightarrow Seq Instr begin \\ Instr &\rightarrow east north west south \end{aligned}$	6 (1+2+3)	CO2	An
i)	Write the language definition of the above grammar G_2 with $\Sigma = \{a, b\}$ and generate the leftmost and rightmost derivation for the input string " $aaab$ ".			
ii)	For a robot position tracking system CFG G_3 , suppose a robot can instructed to move east, north, west, or south from the current position. Position can be marked by a coordinate pair (x, y) with an initial position $(0,0)$ at the beginning. Draw the tracking position of the robot and give the leftmost derivation and parse tree for the input string " $begin west south east east north north$ ".			

Symbol
Table
handles

- b) Describe the roles of lexical analyzer and parser in a compiler with the appropriate block diagram. Identify the tokens, lexemes and patterns of the following C programming code segment in a table [please avoid the redundant entry of tokens]:

4 CO2 U
(2+2)

```
int c, n=5, fact = 1;
printf("Enter a integer number\n");
for(c = 1; c <= n; c++)
    fact = fact * c;
printf("Factorial of %d = %d\n", n, fact);
```

3. a) Write the regular expression and draw the nondeterministic finite automata (NFA) transition diagram to define the number of C programming language including fractional and exponential parts. Construct the ϵ -NFA for regular expression $(0|1)^*011$ and convert it into the deterministic finite automata (DFA) using ϵ -closure.

6 CO2- An
(2+4)

- b) Define the different types of grammar with emphasizing the basic production rules of the grammar. Identify the regular grammars, right-linear grammars, left-linear grammars, context free grammars, context sensitive grammars and useless non-terminals in the following grammars. [Some grammar may fit more than one category].

4 CO2 Ap
(2+2)

G4:	G5:	G6:	G7:
$S \rightarrow baB$	$S \rightarrow aX bY$	$S \rightarrow aA \epsilon$	$S \rightarrow Ma Mb$
$aB \rightarrow aA$	$X \rightarrow aY$	$A \rightarrow Ab a$	$M \rightarrow Nb d$
$A \rightarrow bA b$	$Y \rightarrow b$	$C \rightarrow c$	$N \rightarrow c$

END

$L \rightarrow 01 \dots 19$

$L^+ \cdot L^*$

0. S

0

3

9. S

$\Delta a = \text{id} \Delta (1 \Delta 5) / 9$
 $\Delta S = \text{id} a -$

$\Delta a = \text{id} 5 / 9$
 $\Delta S = \text{id} a = \text{id} 5 + 4$
 $\Delta 6 = \text{id} 3 + \Delta 5$
 $\Delta 2 = \text{id} 2$
 $\text{id} 1 = \text{id} 2 + \Delta 6$

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International Islamic University Chittagong
Center for General Education (CGED)
Midterm Examination, Spring-2025

Leaders

Course Code: URED-3503(URED 2101 for LLB)
Course Title: Political Thought & Social Behavior

Full Marks: 30

Time: 1 hour & 30 minutes

Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question.

	Questions	Marks	CLOs	Bloom's taxonomy domain
1	What is Islamic political system? Explain its fundamental principles.	10	2	Remember & Create
2	Define <i>Shari'ah</i> and explain some characteristics of a constitution under Islamic <i>Shari'ah</i> .	10	2	Remember & Create
3	a) What are the qualities of the chief executive of an Islamic state? Analyze the functions of the executive. Or b) Evaluate the legislative organ of the government from Islamic perspective and briefly explain the function of <i>Majlish-e-Surah</i> .	10	2	Remember & Analyze Analyze & Create

to endorse Law

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